

Use Web Browsers to Test Hybrid Mobile Apps and Mobile Web Apps

Mobile Apps and mobile Web apps, which are the rage today, have to undergo a process of testing before they can be universally used. This article discusses how Web browsers in the form of emulators and simulators can be used to test mobile apps even in the absence of the actual mobile device.



Mobile app testing is a very vast and challenging field, given the diversity of mobile devices today. With each different device, there are many different elements that need to be considered for testing. The common elements comprise things like form factors (screen size and resolution), network operators, the platform, etc. There are even more factors that come into play if the app in consideration is a hybrid mobile app. Mobile Web apps can sustain even without a rigorous testing phase, because they depend on only a few of the factors listed above. These days, every website that is being developed takes into consideration the approach of responsive Web design. If the design is robust enough, it will adapt to mobile browsers on its own and hence, removes the need for testing. On the other hand, hybrid mobile apps have to go through a proper testing phase because what affects them is much more than just the form factor.

Testing should, ideally, be done on actual devices, but as already mentioned, this is a very vast field in itself and there are many organisations solely dedicated to the purpose of mobile testing. This option can be considered in the final testing phases. For initial unit testing and integration testing, especially the testing done by the developer during the process of development, there are two major approaches—simulators and emulators. I would like to differentiate between them because more often than not, people are confused by these terms, often using them interchangeably, which is not correct. In simple terms, simulators are for analysing, and emulators are used as substitutes for the real device. So if you do not have a real device, you can use an emulator, because it mimics the outward behaviour of the device. But the emulation mechanism does not have to correctly imitate the internal state of the target it is emulating. Simulators, on the other hand, try to model the state of the target.